

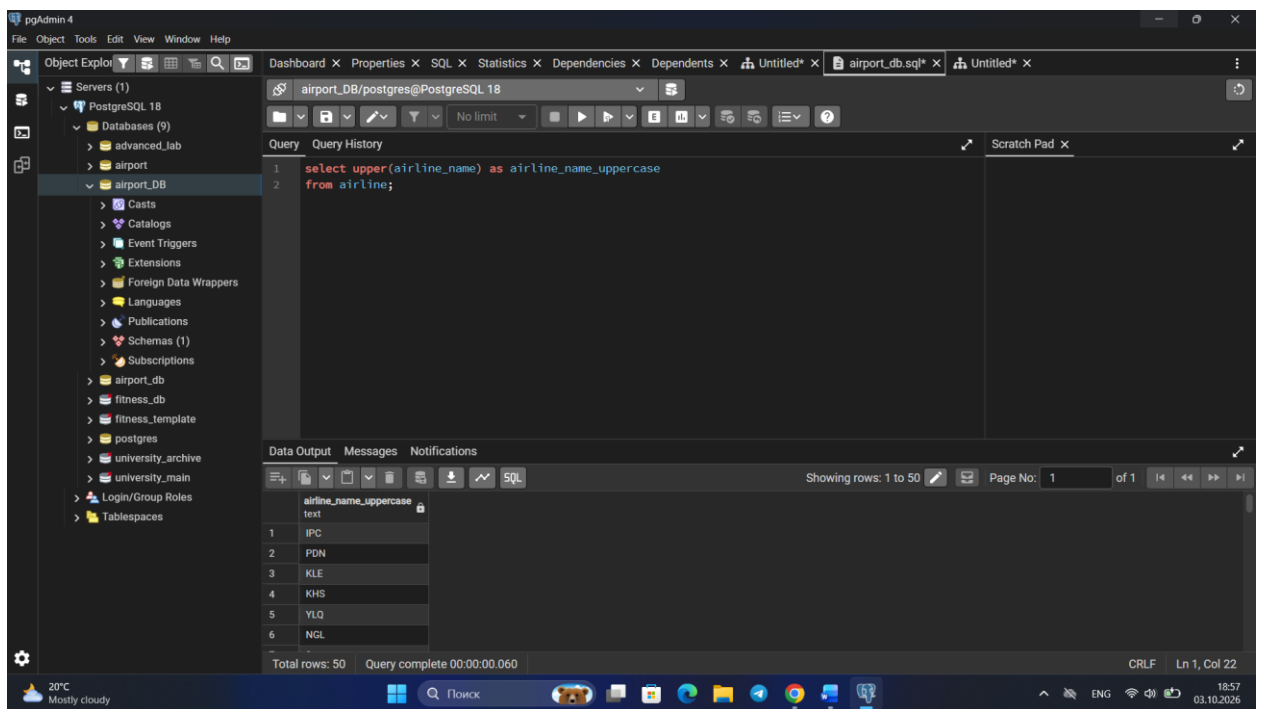
Laboratory work 4

We continue to work with the database from the previous laboratory works.

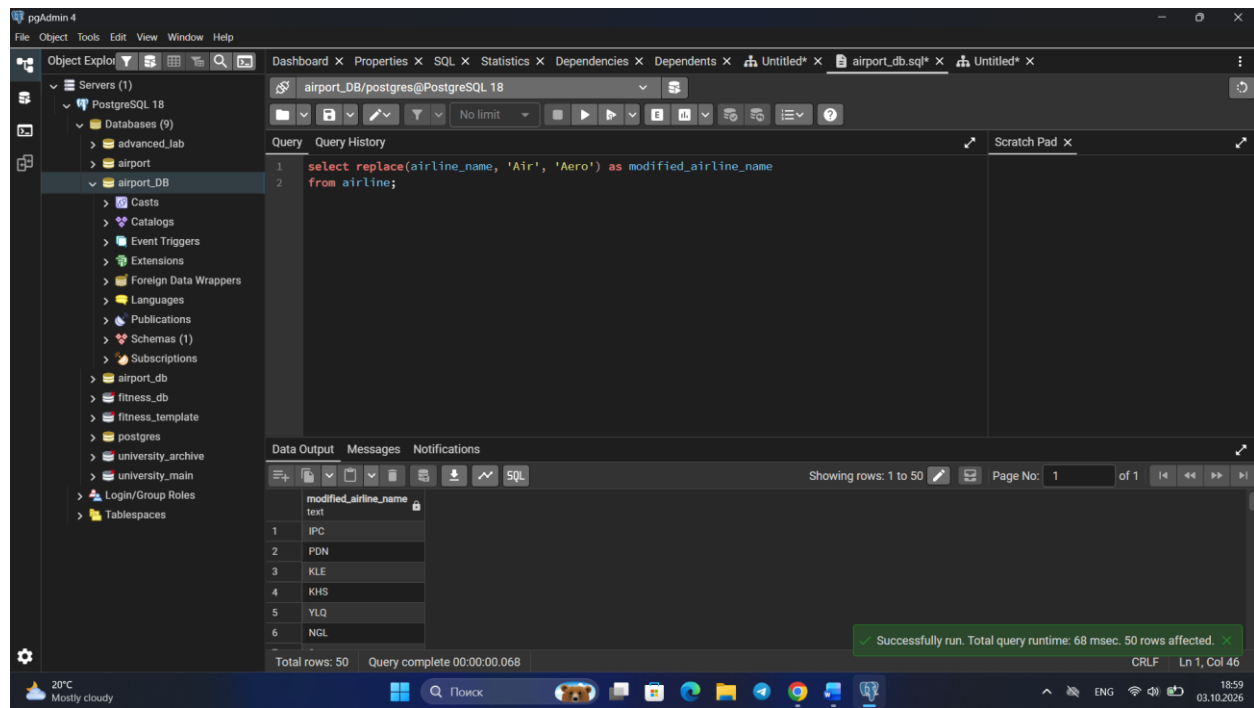
Take a full-page screenshot that covers the code and results of each task.

Tasks:

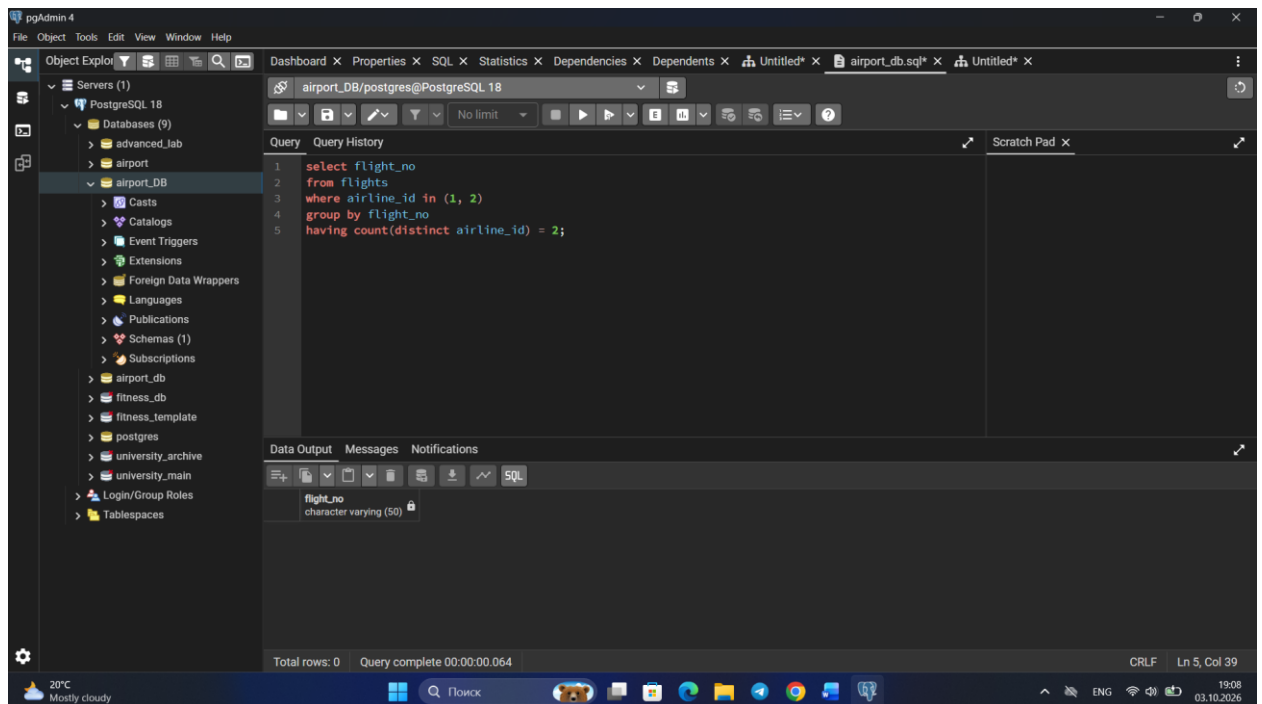
1. Retrieve all airline names in uppercase.



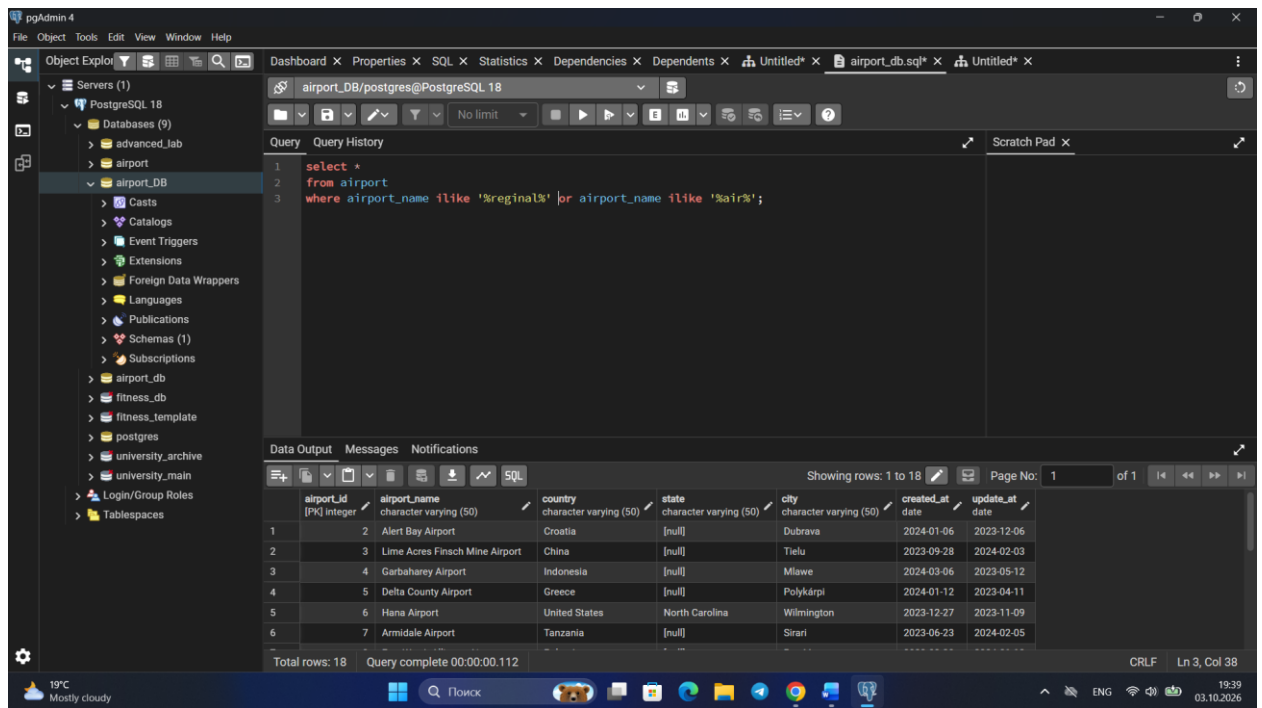
2. Replace any occurrence of the word "Air" in airline names with "Aero".



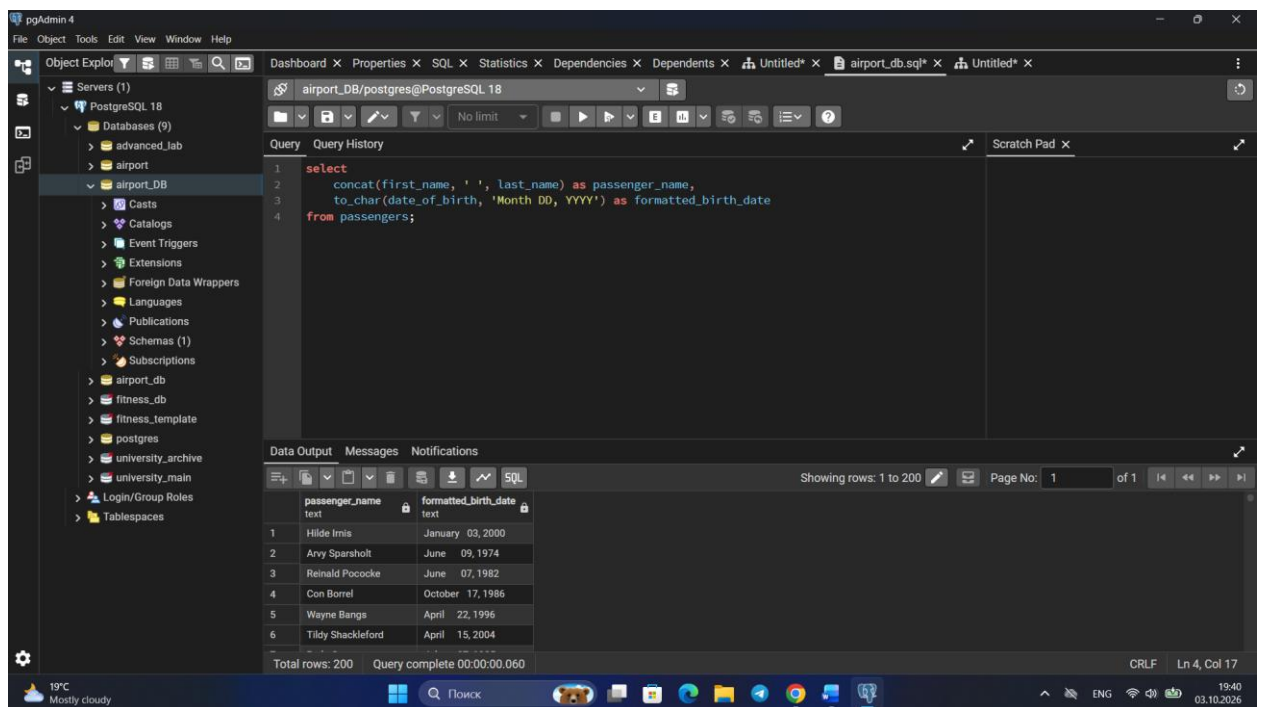
3. Find all flight numbers that coordinates with both airline 1 and airline 2.



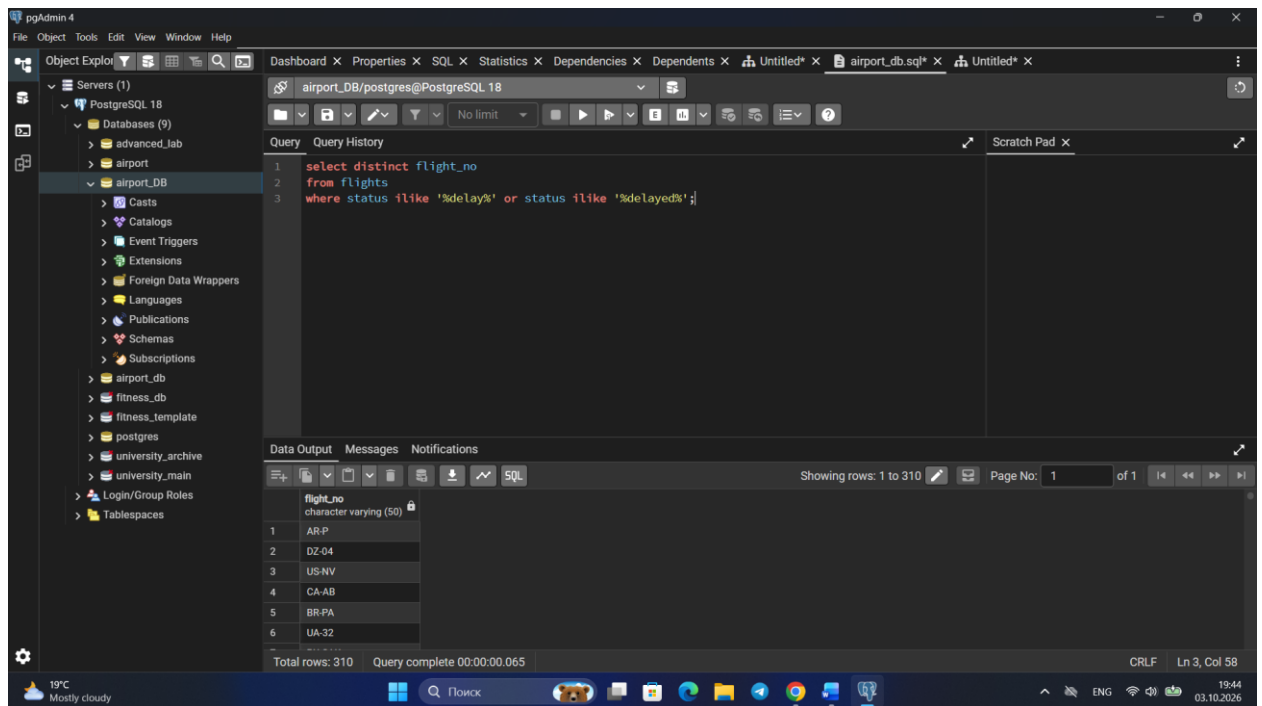
4. Retrieve airports that contain the word "Reginal" and "Air" in their names.



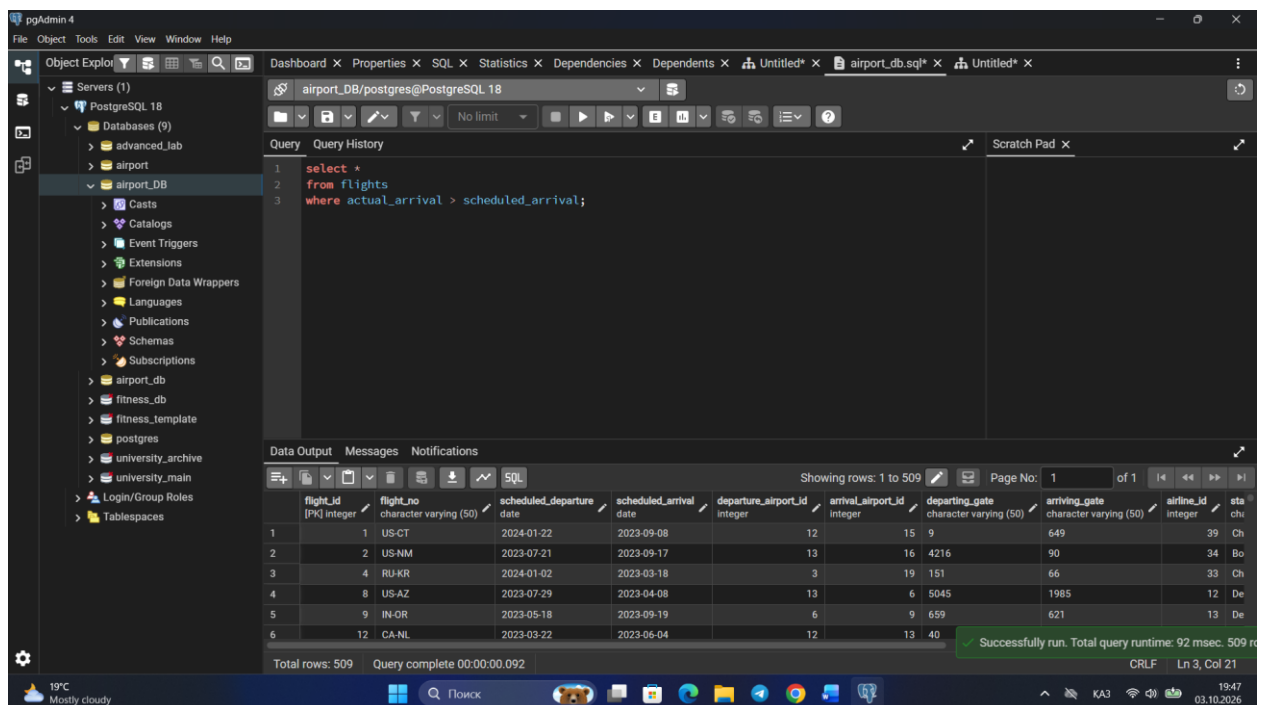
- Retrieve passenger names and format their birth dates as 'Month DD, YYYY'.



- Find flight numbers that have been delayed based on the actual arrival time.



- Find flights that arrived late according to their actual arrival time compared to the scheduled arrival time.



- Show airlines from any of: ('France','Portugal','Poland') created between '2023-11-01' and '2024-03-31'.

pgAdmin 4

Object Explorer: Servers (1) PostgreSQL 18 Databases (9) advanced_lab airport airport_DB Casts Catalogs Event Triggers Extensions Foreign Data Wrappers Languages Publications Schemas Subscriptions airport_db fitness_db fitness_template postgres university_archive university_main Login/Group Roles Tablespaces

Query: airport_DB/postgres@PostgreSQL 18

```

1 select *
2 from airline
3 where airline_country in ('France', 'Portugal', 'Poland')
4 and created_at >= '2023-11-01'
5 and created_at <= '2024-03-31';

```

Data Output: Showing rows: 1 to 7 Page No: 1 of 1

airline_id [PK] integer	airline_code character varying (50)	airline_name character varying (50)	airline_country character varying (50)	created_at date	update_at date
1	4	OOKB	KHS	France	2023-11-18
2	13	OIFE	IFH	Portugal	2024-01-09
3	15	SYKZ	KKG	Poland	2023-12-24
4	24	ZSRG	RUG	Portugal	2024-03-11
5	28	RKPD	JDG	Poland	2023-11-03
6	30	CYST	YST	Poland	2023-11-24

Total rows: 7 Query complete 00:00:00.071 Successfully run. Total query runtime: 71 msec. 7 rows affected.

9. Find top3 overweighted baggage with more than 25kg.

pgAdmin 4

Object Explorer: Servers (1) PostgreSQL 18 Databases (9) advanced_lab airport airport_DB Casts Catalogs Event Triggers Extensions Foreign Data Wrappers Languages Publications Schemas Subscriptions airport_db fitness_db fitness_template postgres university_archive university_main Login/Group Roles Tablespaces

Query: airport_DB/postgres@PostgreSQL 18

```

1 select *
2 from baggage
3 where weight_in_kg > 25
4 order by weight_in_kg desc
5 limit 3;

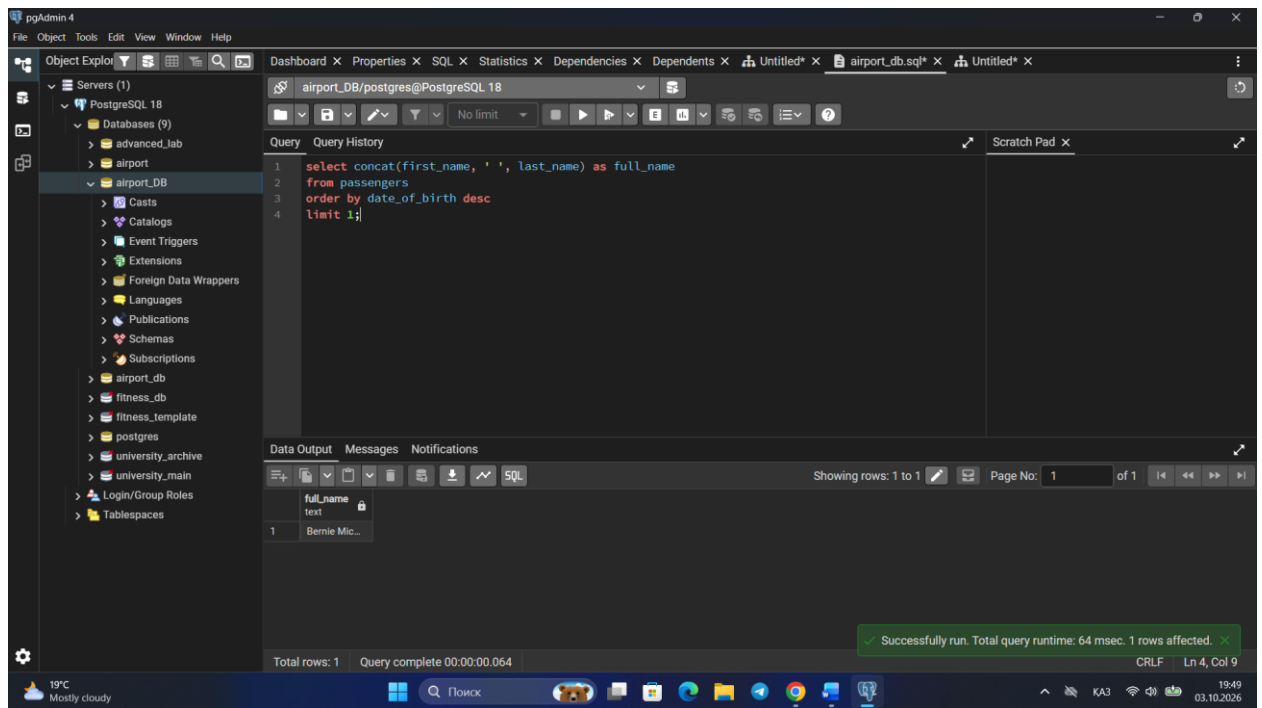
```

Data Output: Showing rows: 1 to 3 Page No: 1 of 1

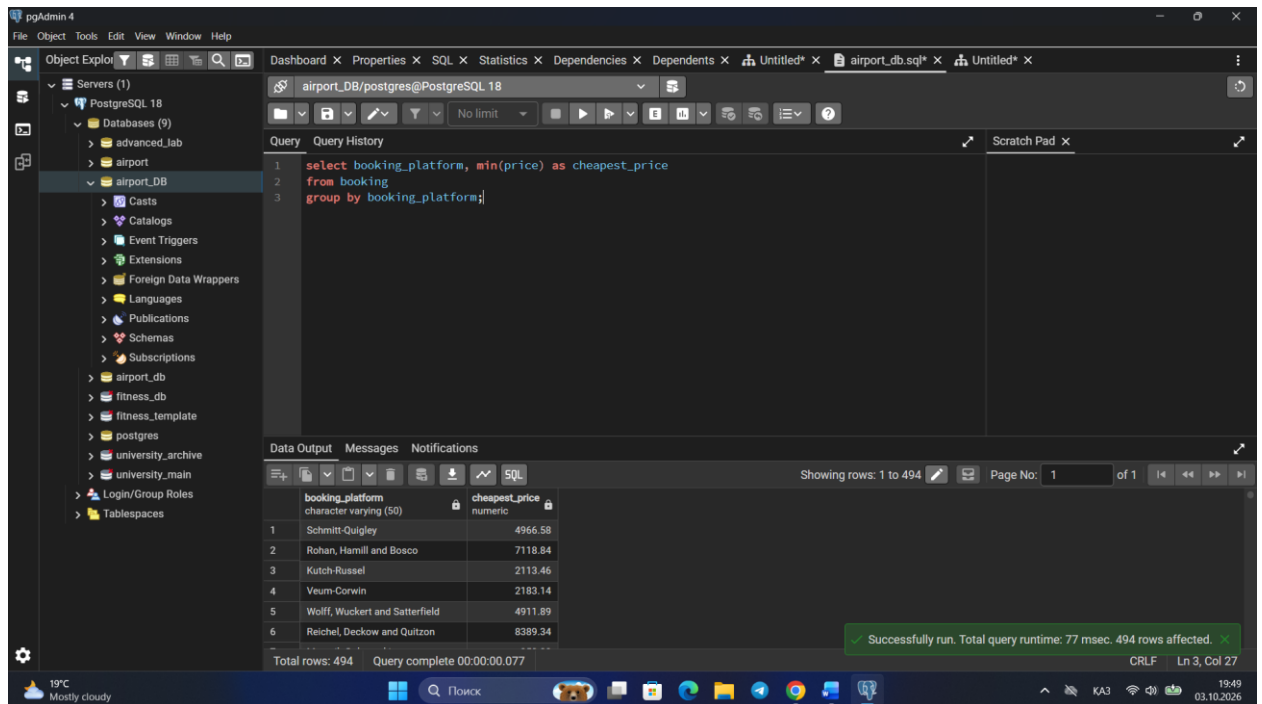
baggage_id [PK] integer	weight_in_kg numeric (4,2)	created_date date	update_date date	booking_id integer	
1	321	49.80	2023-07-07	2023-11-10	349
2	61	49.57	2023-04-21	2023-07-31	464
3	102	49.44	2024-01-24	2023-11-01	53

Total rows: 3 Query complete 00:00:00.141 Successfully run. Total query runtime: 141 msec. 3 rows affected.

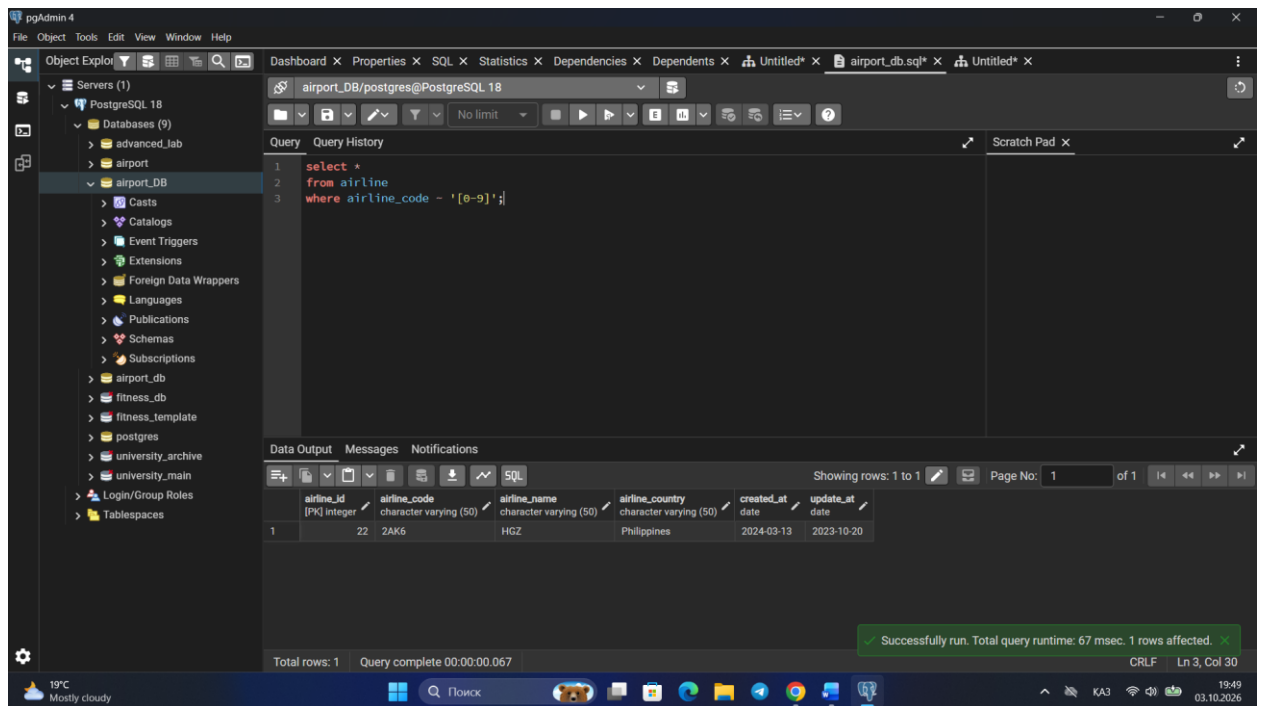
10. Find the youngest passengers' full name.



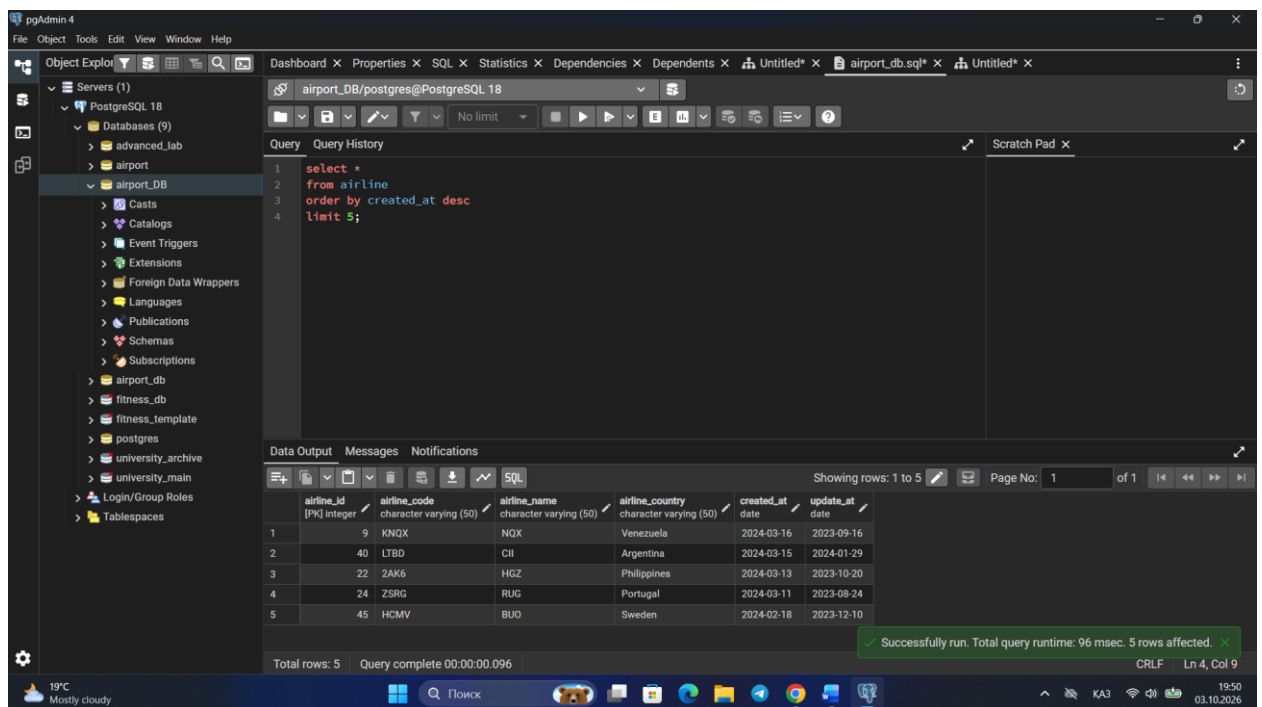
11. Find the cheapest booking price on each booking platform.



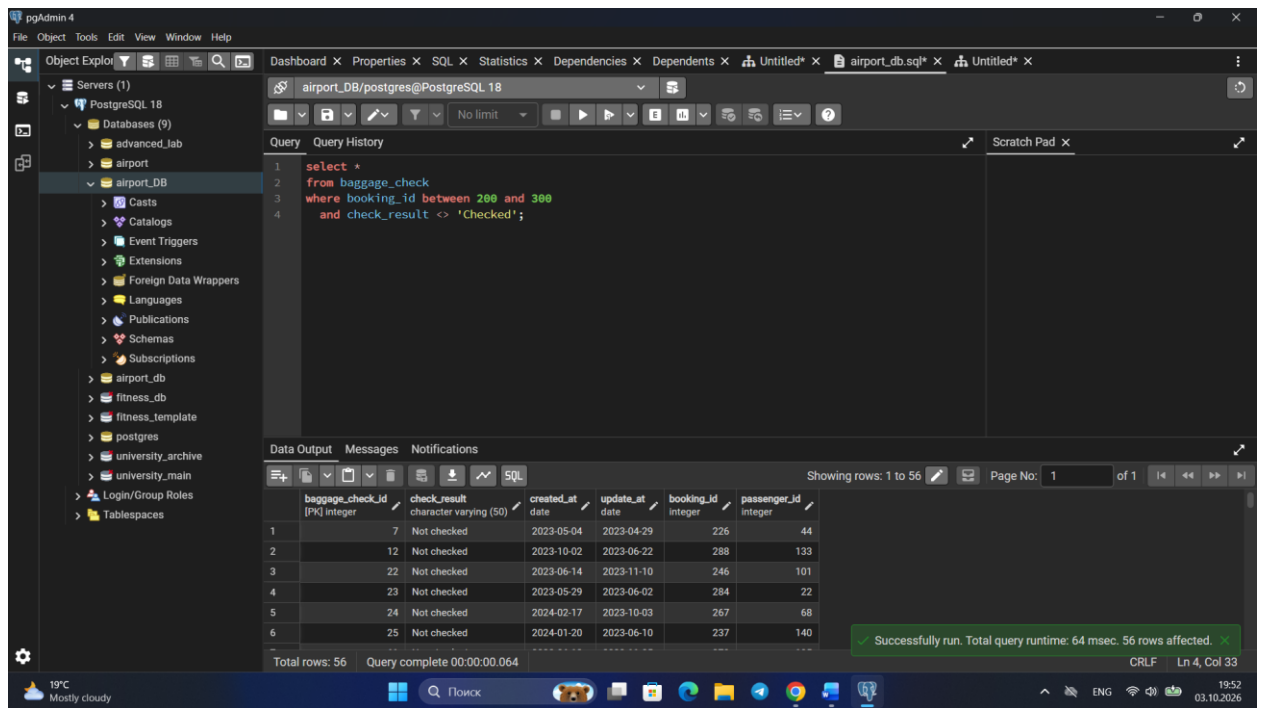
12. Return airlines whose airline_code contains a digit.



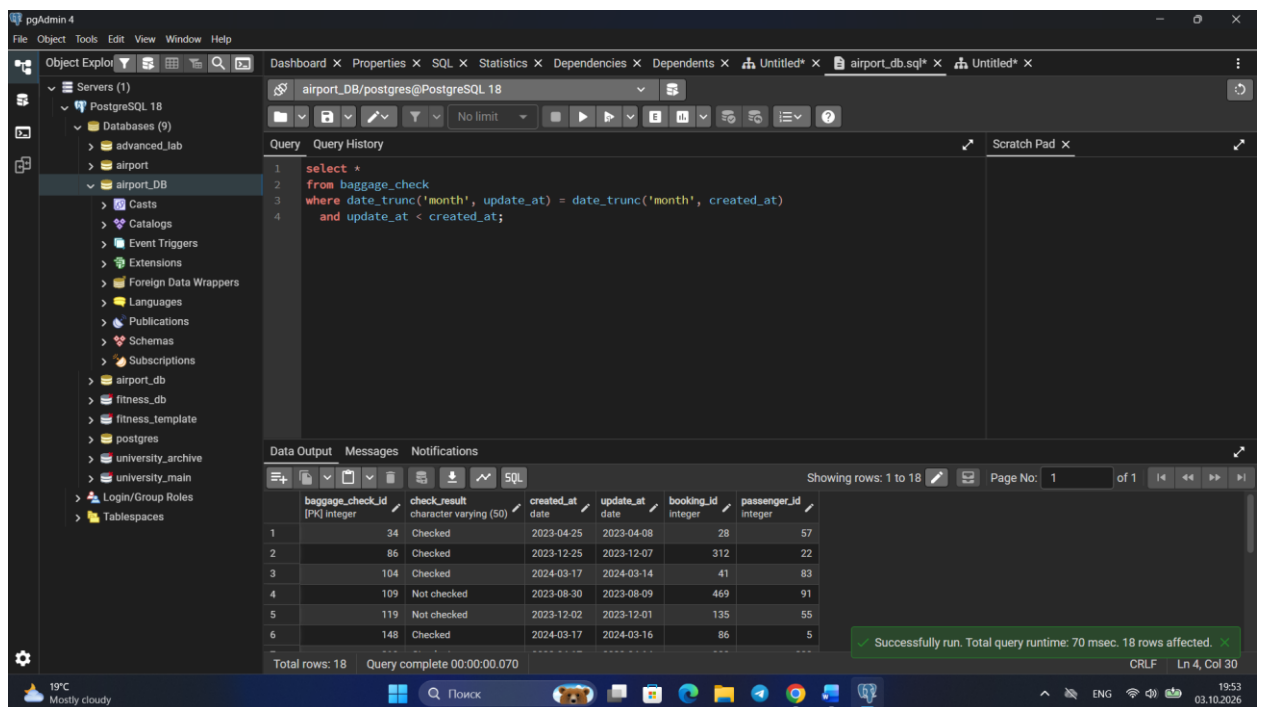
13. List the top 5 most recently created airlines.



14. Return all rows where booking_id is between 200 and 300 inclusive and check_result <> 'Checked'.



15. Baggage checks where `update_at` is in the same month as `created_at` but occurs earlier than `created_at`.



Conclusion:

In this laboratory work, various SQL queries were executed to perform data manipulation, filtering, and aggregation on the database. PostgreSQL functions such as string, date, pattern matching, and conditional filtering were applied to solve practical database problems.